UNIT – II

Machines and Measuring Instruments:

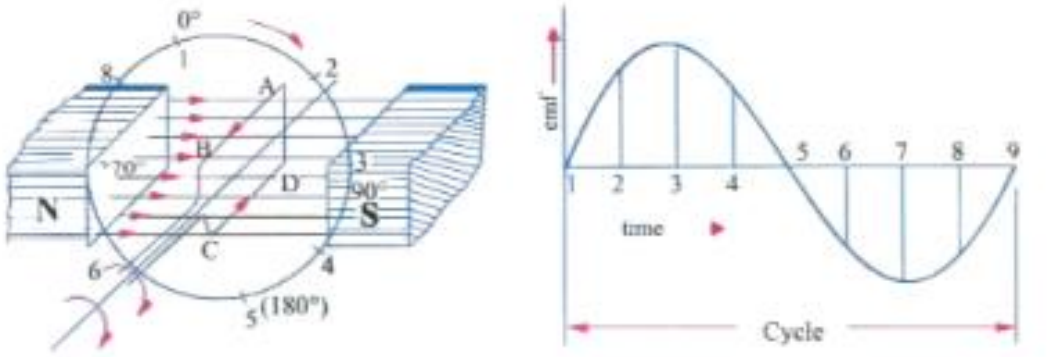
Principles and operation of DC machines, Transformers – Synchronous Machines - three Phase and single phase induction motors - Moving coil and moving iron instruments, Wheatstone bridge and Megger.

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1. Working Principle of DC Generators

An electrical generator is a machine which converts mechanical energy (or power) into electrical energy (or power). whenever a conductor cuts magnetic flux, dynamically induced e.m.f. is produced in it according to Faraday's Laws of Electromagnetic Induction. This e.m.f. causes a current to flow if the conductor circuit is closed. Hence, two basic essential parts of an electrical generator are (i) a magnetic field and (ii) a conductor or conductors which can so move as to cut the flux.

In Fig. is shown a single-turn rectangular copper coil ABCD rotating about its own axis in a magnetic field provided by either permanent magnets or electromagnets. The two ends of the coil are joined to two slip-rings 'a' and 'b' which are insulated from each other and from the central shaft. Two collecting brushes (of carbon or copper) press against the slip-rings. Their function is to collect the current induced in the coil and to convey it to the external load resistance R. The rotating coil may be called 'armature' and the magnets as 'field magnets'.



Consider a single loop DC generator (as shown in the figure), in this a single turn loop 'ABCD' is rotating clockwise in a uniform magnetic field with a constant speed. When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously. This change in flux linkage induces an EMF in coil sides and the induced EMF in one coil side adds the induced EMF in the other.

The EMF induced in a DC generator can be explained as follows

- When the loop is in position-1, the generated EMF is zero because, the movement of coil sides is parallel to the magnetic flux.
- When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a small EMF is generated.
- When the loop is in position-3, the coil sides are moving at right angle to the magnetic flux, therefore the generated EMF is maximum.
- When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a reduced EMF is generated in the coil sides.
- When the loop is in position-5, no flux linkage with the coil side and are moving parallel to the magnetic flux. Therefore, no EMF is generated in the coil.
- At the position-6, the coil sides move under a pole of opposite polarity and hence the polarity of generated EMF is reversed. The maximum EMF will generate in this direction at position-7 and zero when at position-1. This cycle repeats with revolution of the coil.

It is clear that the generated EMF in the loop is alternating one. It is because any coil side (say AB) has EMF in one direction when under the influence of N-pole and in the other direction when under the influence of S-pole. Hence, when a load is connected across the terminals of the generator, an alternating current will flow through it. Now, by using a commutator, this alternating emf generated in the loop can be converted into direct voltage. We then have a DC generator.

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2. Working Principle of DC Motor

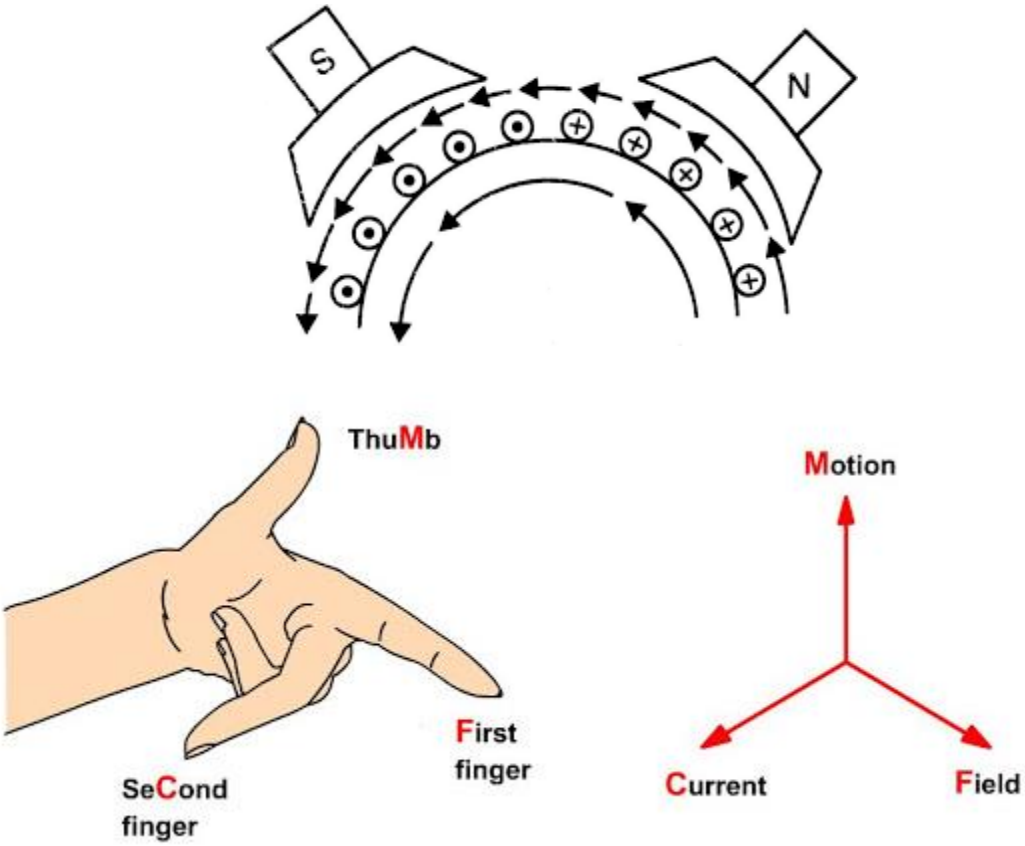
An Electric motor is a machine which converts electric energy into mechanical energy. Its action is based on the principle that when a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force whose direction is given by Fleming's Left-hand Rule and whose magnitude is given by F = BIL newton

Where,

B = magnetic flux density,

I = current and

L = length of the conductor within the magnetic field.

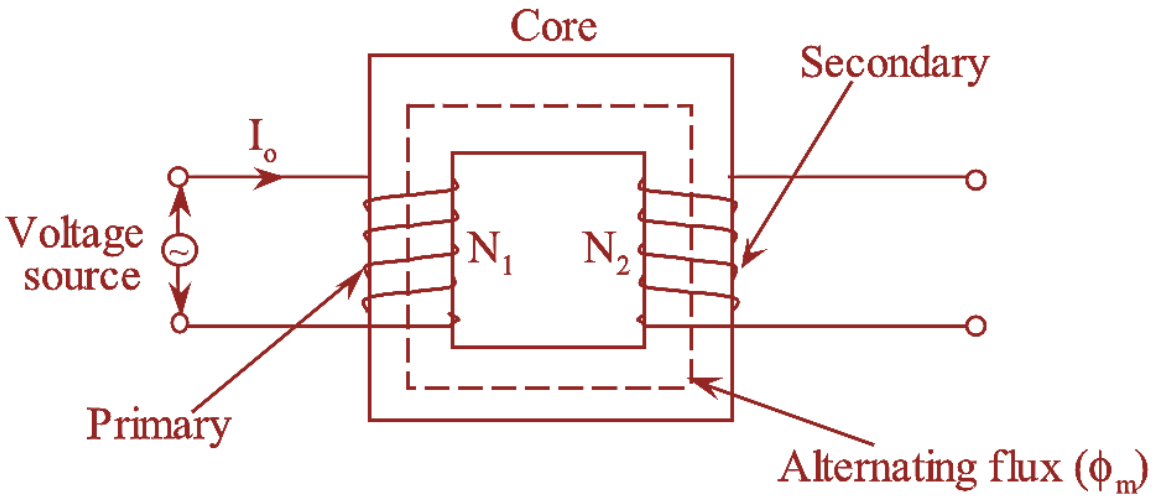


In Fig. is shown a part of multipolar d.c. motor. When its field magnets are excited and its armature conductors are supplied with current from the supply mains, they experience a force tending to rotate the armature. Armature conductors under N-pole are assumed to carry current downwards (crosses) and those under S-poles, to carry current upwards (dots). By applying Fleming's Left-hand Rule, the direction of the force on each conductor can be found. It is shown by small arrows placed above each conductor. It will be seen that each conductor experiences a force which tends to rotate the armature in anticlockwise direction. These forces collectively produce a driving torque which sets the armature rotating.

It should be noted that the function of a commutator in the motor is the same as in a generator. By reversing current in each conductor as it passes from one pole to another, it helps to develop a continuous and unidirectional torque.

3.Working Principle of Transformer

A transformer is a stationary electric machine which transfers electrical energy (power) from one circuit to another circuit with the same frequency but with different voltages and currents.



- The transformer works on the principle of Mutual induction between two magnetically coupled coils is based on Faraday's laws of Electromagnetic Induction
- When primary winding is connected to AC supply mains a current flow through it
- This current produces an alternating flux in the core
- This flux links with primary winding and produces self-induced EMF in the primary winding which opposes the applied voltage
- This flux passes through the core and links with the secondary windings to induce an E.M.F called mutually induced E.M.F in the secondary winding
- The frequency of the EMF induced in the secondary is same as that of the flux or that of the supply voltage.
- The induced EMF in the secondary winding will be able to circulate the current in the external load connected to it
- Energy is transferred from primary winding to the secondary winding by electro-magnetic induction principle without change of frequency

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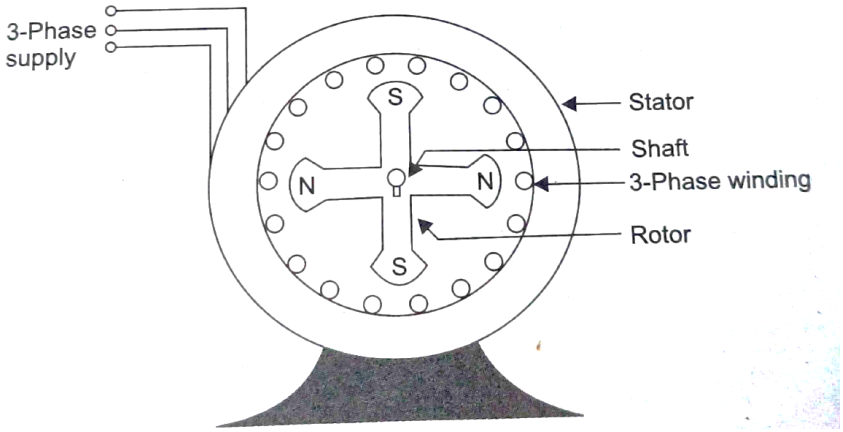
4. Synchronous Machine

❖ Based on the energy conversion, synchronous machines may be classified into two types −

1. Synchronous Generator
2. Synchronous Motor

- The synchronous machines are the most extensively used electrical machines in power system applications like **power generator, power factor correction, driving constant speed mechanical load**, etc.
- The synchronous machine which converts mechanical energy into alternating current electricity is called a **synchronous generator or alternator**. While the synchronous machine which converts alternating current electricity into mechanical energy is called a **synchronous motor**.

- An alternator is an electrical machine which converts mechanical energy into alternating electric energy.
- The working principle of alternator is based on the **Faraday law of electromagnetic induction**, which says that whenever a conductor cuts the magnetic flux an e.m.f induced in it when there is a relative motion between that conductor an magnetic field.
- The rotor winding is energized from the DC exciter and hence alternatively N and S poles are developed in the rotor.
- When the rotor is rotated in the anti-clockwise direction by a prime mover, the stator or armature conductors are cut by the magnetic flux of rotor poles. Consequently, an e.m.f. is induced in the armature conductors due to electromagnetic induction.



- The induced e.m.f. is alternating nature, since N and S poles of rotor alternate pass the armature conductors. The direction of induced e.m.f. can be found by Fleming righs hand rule and frequency is given by f = PN / 120 where N = speed of rotor in r.p.m. and number of rotor poles.
- Synchronous speed is given by ,$N_s = \frac{120\,f}{P}$

  Where f= frequency of the generated e.m.f.

  P=No.of poles

- The magnitude voltage induced in each phase depends upon the rotor flux, the number and position of the conductors and the speed of the rotor.
- When the rotor is rotated, a 3-phase voltage is induced in the armature winding. The magnitude of induced e.m.f. depends upon the speed of the rotation and the DC exciting current. The magnitude of e.m.f. in each phase of the armature winding is the same, bu differs in phase angle by 120 deg electrical.

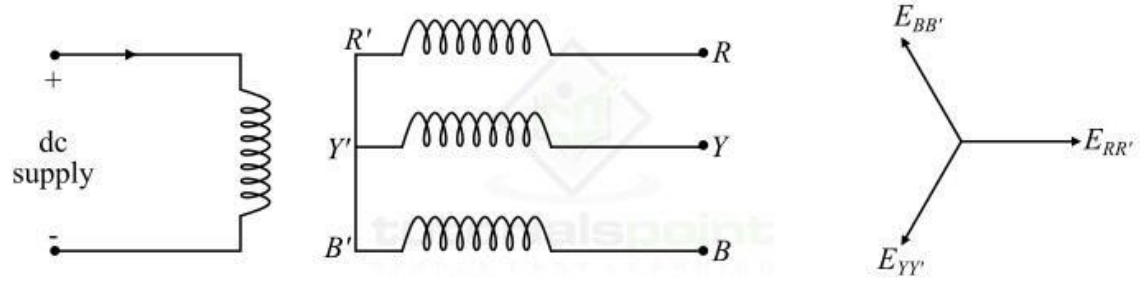
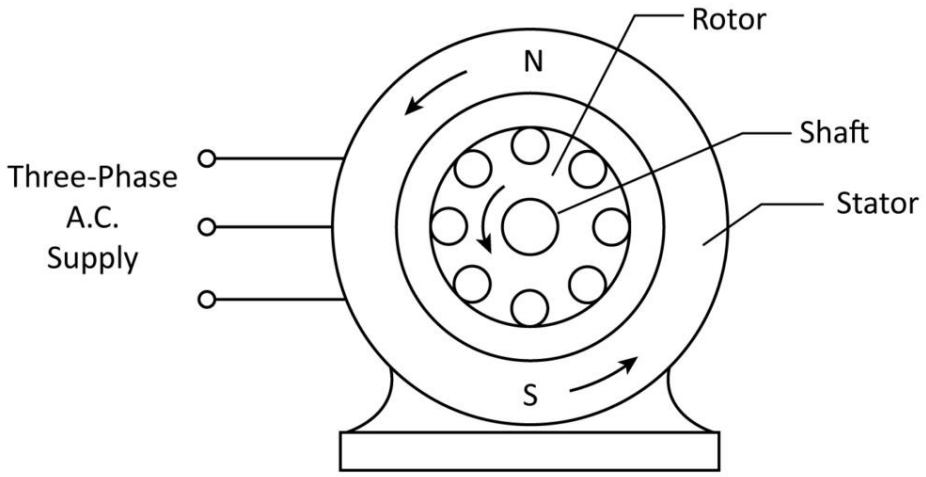


Figure 1 - Electrical Equivalent Circuit & Phasor Diagram

5. Three phase induction motor

- An electric motor converts electrical energy into a mechanical energy, which is then supplied to different types of loads.
- AC motors operate on an AC supply, and they are classified into synchronous motors, single phase and 3-phase induction and special purpose motor.
- Out of all types, 3-phase induction motors are most widely used for industrial applications manly, because they have simple and rugged construction, low cost, high efficiency, self-starting torque and low maintenance.
- A 3-phase induction motor is a single-excited AC machine. Its stator winding is directly connected to AC source and its rotor winding receives its energy from stator by means of connected exactly in the same as the secondary of a two-winding transformer receives of its induct from the primary. An induction motor derives its name from the fact that the rotor current is induced by the magnetic field, instead of electrical connections.

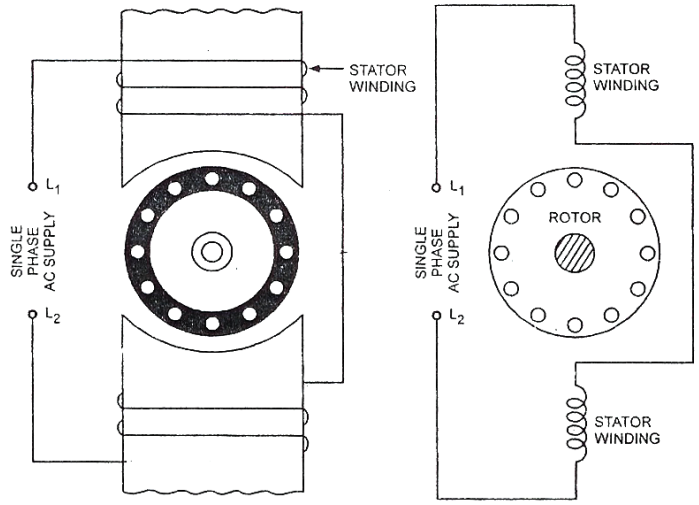


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6. Single phase induction motor

- The single phase induction motor, also known as the **asynchronous motor**, is a commonly used type of electric motor that can operate using a single-phase power supply.

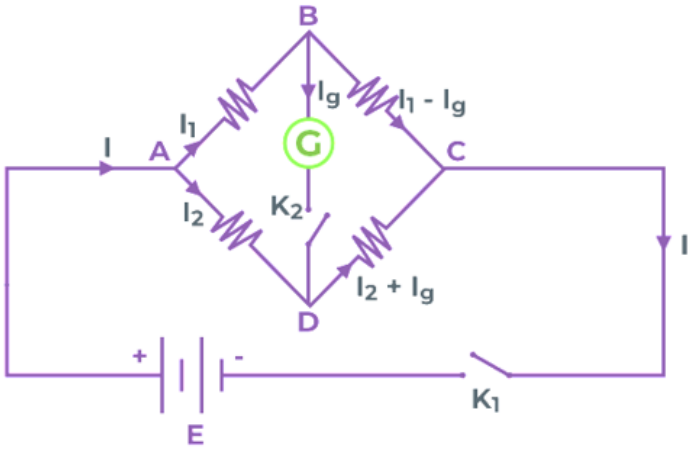
- The single phase induction motor remains one of the most widely used motors for **domestic, commercial, and small industrial applications** where only single-phase power is available.
- The single phase induction motor stands out as one of the most ubiquitous and essential devices.
- This type of motor is widely used in various **household and industrial applications** due to its simplicity.
- A single-phase induction motor is similar in construction to that of a polyphase induction motor with the difference that its stator has only one winding.
- When a single phase AC supply is applied to the stator winding of single phase induction motor, the alternating current starts flowing through the stator or main winding.
- The flux is then generated by the AC current. The flux also links with the rotor conductors and hence cut the conductors of the rotors.
- According to the Faraday's law of electromagnetic induction, emf gets induced in the rotor.
- The current begins to flow in the rotor after the rotor circuit is closed.
- The flux created by this rotor current is known as the rotor flux.
- Since this flux is produced due to the induction principle so, the motor working on this principle got its name as an induction motor.



- There are currently two fluxes: the main flux and the rotor flux.
- The desired torque, which the motor needs to rotate, is produced by these two fluxes.
- Its motors consist of a single-phase winding mounted on the stator and a cage winding on the rotor.
- When a single-phase supply connected to stator winding pulsating magnetic field is produced.
- Under these conditions, the rotor does not rotate. It requires some special starting methods.

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7. Wheatstone Bridge



- Wheatstone Bridge is an instrument designed to measure unknown resistance in electrical circuits.
- It calculates the unknown resistance by balancing the two legs of the bridge circuit where one leg contains both known resistors and the other leg contains one known (variable) and one unknown resistor.
- Wheatstone Bridge works on the principle of null deflection i.e., there is no current flowing through the galvanometer, and its needle shows no deflection, hence the name null deflection.
- In the unbalanced state of the Wheatstone bridge i.e., when the potential across the galvanometer is different, the galvanometer shows the deflection, and as the bridge

becomes balanced by changing the variable resistor, the potential difference across the galvanometer becomes zero i.e., the equilibrium state of Wheatstone bridge.

- Construction of Wheatstone Bridge requires four resistors P, Q, R, and S that are placed in the form of four sides AB, BC, AD, and DC of a quadrilateral ABCD. A cell E and key K1 are placed between the A and C ends of this quadrilateral, and a sensitive galvanometer G and key K2 is placed between the B and D ends. Clearly, the potential of point A will be equal to the potential of the positive plate of the cell and the potential of point C will be equal to the potential of the negative plate of the cell.
- It is clear from the figure that the resistances P and Q are in series when the key K2 is open. Similarly, resistances R and S are in series, but P and Q together (arm ABC) and R and S together (arm ADC) are connected in parallel to each other. Since the side BD of the galvanometer is placed like a bridge over the sides ABC and ADC of the quadrilateral, this circuit is called a bridge.
- When the bridge is in an equilibrium state, that is, there is no deflection in the galvanometer. That is, in the equilibrium state of the bridge, the ratio of the resistances of any two adjacent arms is equal to the ratio of the remaining two adjacent sides.
- Suppose, on pressing the cell key K_1, a current I flows through the cell, which splits into two parts at the end A.
- One part I_1 flows through the resistance P in arm AB and the other part I_2, through the resistance R in arm AD.
- The current I_1 again comes to end B and gets divided into two parts.
- One part of it Ig flows through the galvanometer in arm BD and the remaining part $(I_1 – I_g)$ flows through resistance Q in arm BC.
- At the end D, the current I_2 from arm AD and the current I_g from arm BD, so the current flowing through the resistance S in arm DC will be $(I_2 + I_g)$.
- So according to Kirchhoff's law, in closed path ABDA,

$$I_1P + I_gG – I_2R = 0 \qquad \ldots (1)$$

- And in closed path BCDB,

$$(I_1 – I_g)Q – (I_2 + I_g)S – I_gG = 0 \qquad \ldots (2)$$

- The values of the resistors P, Q, R, and S are taken in such a way that no current flows through the galvanometer G when the key K_2, is pressed.
- This is called the equilibrium state of the bridge, that is, in the equilibrium state of the bridge, the deflection in the galvanometer is zero ($I_g = 0$).
- Putting Ig = 0, in the above equations,

  $I_1P = I_2R$ and $I_1Q = I_2S$ or $I_1P / I_1Q = I_2R / I_2S$ or $P / Q = R / S$
- This is the necessary condition for the balance of the Wheatstone Bridge. With the help of the above formula, knowing the values of three resistors P, Q, and R, the value of the fourth resistance S can be found.

The Wheatstone Bridge Formula for the calculation of the unknown resistor is as follows:

$$R=P/S * Q \text{ OR } R = PS /Q$$

where,
P and Q is the resistance of ratio arm
S is the known resistance of the standard arm
R is the unknown resistance

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8. **Moving iron (MI) type**

- The most common type of ammeters and voltmeters for laboratory or switch board use at power frequencies are the moving iron instruments, because these are very cheap, accurate and simple in construction.

Moving iron instruments are classified into two types:

(i) Attraction type

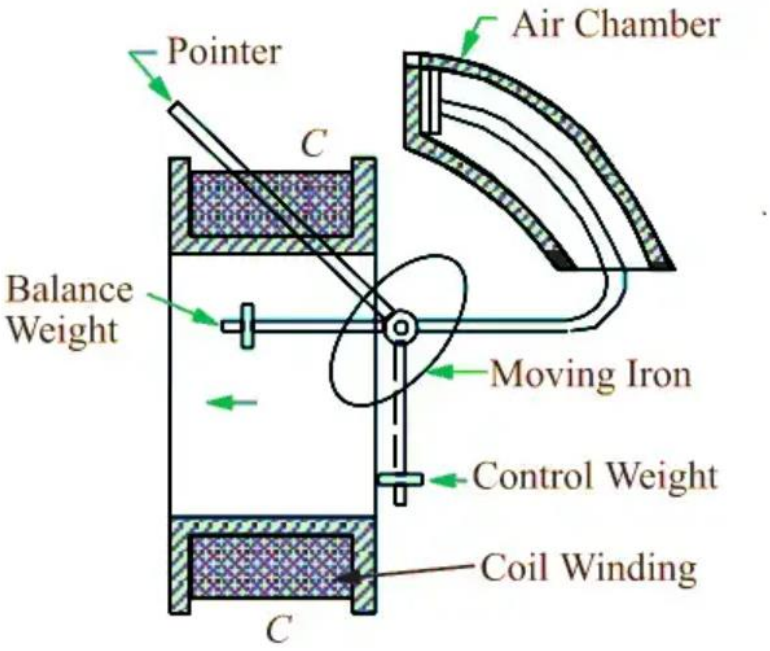
(ii) Repulsion type

I. Attraction Type Moving Iron Instruments

Principle: These instruments are based on the principle that when an unmagnified soft iron piece is placed near the current carrying coil (conductor), then the iron piece is attracted towards the coil. Damping torque is provided by air friction.

Construction: Figure shows the constructional details of attraction type moving iron instrument. It mainly consists of a hollow cylindrical coil or solenoid. An oval shaped soft iron piece and a pointer are attached to the spindle, which is supported between two

jeweled bearings near the coil. The soft iron piece is free to move in or out of the coil, and then the pointer also deflects on the scale with the motion of the iron piece. This is provided with spring control and air friction damping.



Working:

- When the instrument is connected to the circuit, the operating current flows through the stationary coil.
- A magnetic field is set up and the soft iron piece is magnetized, which is attracted towards the centre of the coil.
- Thus the pointer attached to the spindle is deflected over the calibrated scale.
- If the current in the coil is reversed, the direction of the magnetic field produced will reverse and the magnetism produced in the soft iron piece will also reverse.
- Hence, the direction of the deflecting torque remains unchanged.
- These instruments can, thus, be used on DC as well as AC systems.
- **Deflection torque :**
- $T_d \propto I^2$
- Controlling torque, $T_c \propto \theta$
- Under equilibrium condition $T_d = T_c$; $\theta \propto I^2$

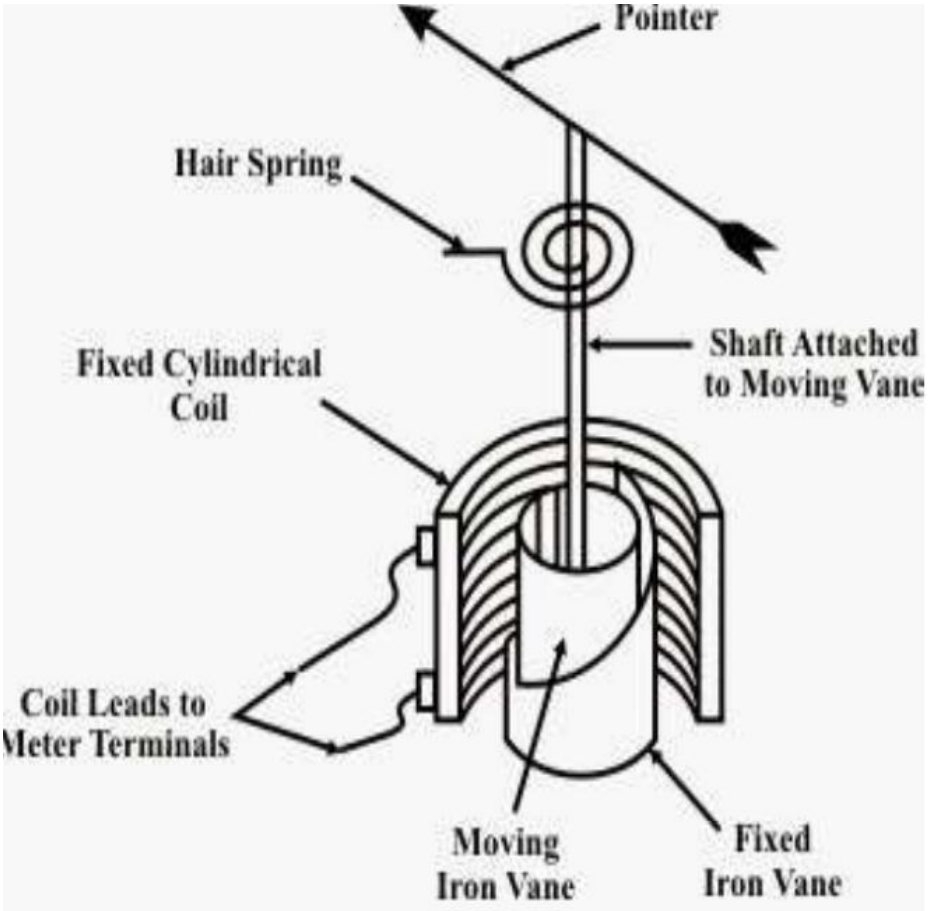
II. Repulsion Type Moving Iron Instruments

Principle:

- These instruments are based on the principle that the repulsive force will act ,when two similarly magnetized iron pieces are placed nearer to each other .

Construction:

- It consists of a fixed cylindrical hollow coil, which carries the operating current.
- Inside the coil there are two vanes placed parallel to each other, one fixed and the other movable, connected to the spindle.
- In this instrument ,controlling torque is provided by spring control method and damping torque is provided by air control damping



Working:

- If current flows through the coil, a magnetic field is set up along the axis of the coil.
- This field magnetizes both the vanes and attains similar polarity.
- A force of repulsion acts between the two vanes.
- Therefore movable vane moves away from the fixed vane.
- The pointer attached to the spindle deflects over the calibrated scale.

Deflecting Torque: The repulsion force between two similarly magnetized iron piece responsible for the deflecting torque.

Since deflecting torque $\propto$ repulsive force

$T_d \propto F$

$T_d \propto I^2$

Controlling torque, $T_c \propto \theta$

Under equilibrium condition $T_d = T_c$; $\theta \propto I^2$

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9. MOVING COIL (MC) INSTRUMENTS

Permanent magnet moving coil (PMMC) instrument:

Principle: It is based on the principle that, whenever a current carrying conductor is placed in a magnetic field, a mechanical force acts on the conductor.

Construction: A PMMC instrument consists of a permanent magnet and a rectangular coil, which can move about a central axis in the air gap, in between the poles N and S of the magnet as shows in figure . The coil consists of a number of turns of copper wire and it is mounted on Aluminimum former and the rectangular coil lies in the narrow air gaps between the poles and cylinder. The coil is mounted on the spindle and acts as a moving element. The purpose of the aluminium former is to provide the necessary damping torque.

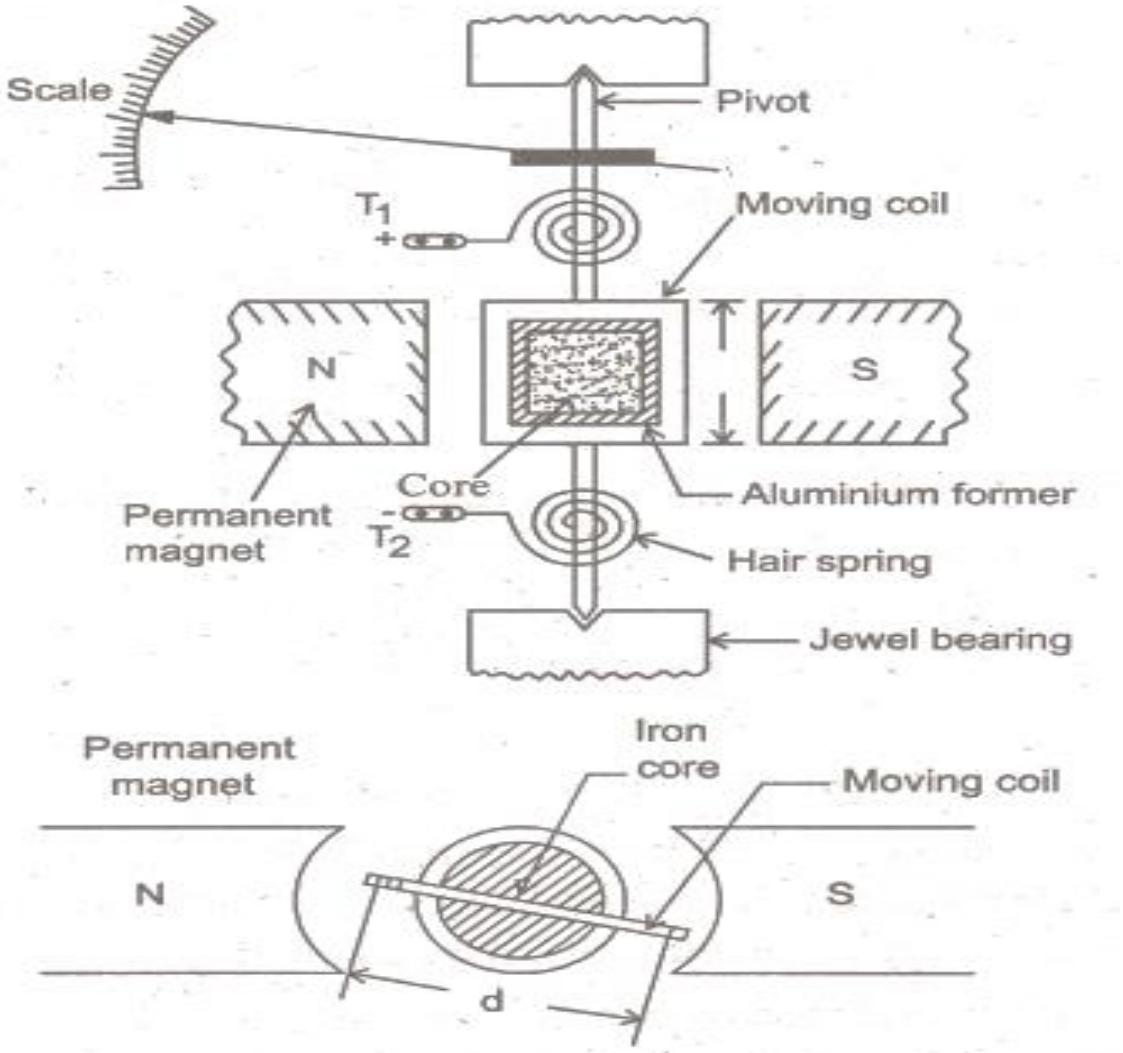


Figure 5.6 PMMC

Working:

When the instrument is connected in the circuit, operating current flows through the coil, which is mounted on the spindle. Since the coil is placed in the strong magnetic field, a force acts on the coil which produces deflecting torque (T_d). Thus pointer attached to the spindle deflected over the scale. If the current in the coil is reversed, the direction of the deflecting torque will be reversed, because field produced by the permanent magnet does not change. This change will give wrong direction of rotation. So this instrument can not be used for AC supply.

10. Megger

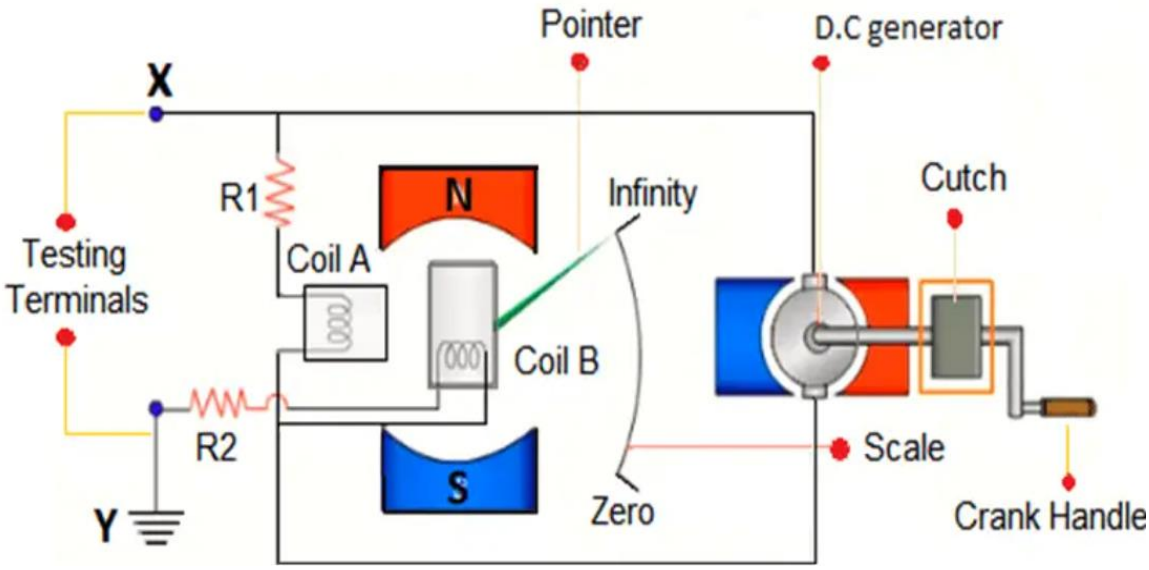
- An instrument that is used to measure insulation resistance is a Megger. It is also known as meg-ohm-meter. It is used in several areas like multi-meters, transformers, electrical wiring, Etc. Megger device is used since the 1920s for testing various electrical devices which can measure greater than 1000meg-ohms.

Insulation Resistance

- Insulation resistance is resistance in ohms of wires, cables, and electrical equipment, which is used to safeguard the electrical systems like electrical motors from any accidental damages like electrical shocks or sudden discharges of current leakages in wires.
- The principle of Megger is based on moving coil in the instrument. When current is flowing in a conductor, which is placed in a magnetic field, it experiences a torque.
- Where vectored Force = strength and direction of the current and magnetic field.
- Case (i) Resistance of insulation = High; pointer of moving coil = infinity,
- Case (ii) Resistance of insulation = Low; pointer of moving coil = zero.
- It is the comparison between Insulation resistance and the known value of resistance.
- It provides the highest accuracy in measurement than other electrical measuring instruments.
- Megger is used to measure a high value of resistance. Megger consists of the following parts.

1. DC generator
2. 2 Coils (Coil A, Coil B)
3. Clutch
4. Crank handle
5. terminal X & Y
6. The megger consists of a DC generator, its armature is hand-driven to generate a voltage. A clutch mechanism is used to slip after reaching a certain determined speed.

7. Resistance R1, R2 connected in series with two coils, Coil A and Coil B which constitute one instrument to which an indicating pointer is attached.
8. Permanent magnets having North, South poles produce a magnetic field to deflect the pointer. Test terminals at X and Y are used to measure insulation resistance.



- The megger is designed in such a way that the needle floats freely till the generator is operated. When the generator is not operating, the needle may come to rest at any point on its scale.
- The megger is used to measure a high quantity of insulation resistance. The high resistance may be between the windings of a transformer or motor or between the conductor in a cable and the conduit or sheath enclosing the cable.
- If the test leads connected to the line and earth terminals are open-circuited and a hand-cranked generator is operated, the needle moves to infinity. Infinity resistance means that it is too high for the instrument to measure.
- If the test leads are connected to each other while the hand crank is turned, the pointer will defect to zero, indicating no resistance between the test leads.
- A zero deflection in the above-mentioned test can mean that the conductor under test is touching the sheath or conduit surround it.

THE END